Hierarchical Deep Learning for Automated CTD Document Classification in Pharmaceutical Regulatory Submissions

CS230 Final Project Report

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Abstract

Pharmaceutical regulatory submissions require organizing hundreds of source documents into a standardized Common Technical Document (CTD) structure. Current approaches rely on keyword matching, achieving only 68% accuracy and requiring extensive manual review. We propose a hierarchical classification approach using BioBERT embeddings that predicts document-to-section mappings at multiple granularity levels (Module → Section). Our model achieves 74.9% accuracy on fine-grained section classification and 98.4% on module-level classification, representing a 7.1 percentage point improvement over baselines.

1. Introduction

The pharmaceutical industry faces a significant challenge in regulatory submissions: organizing thousands of source documents into the ICH Common Technical Document (CTD) format. This process currently requires weeks of manual effort from regulatory professionals, with costs ranging from $500K to $2M per submission.

The Problem: When generating regulatory dossiers using Large Language Models (LLMs), the system must know which source documents contain relevant information for each CTD section. Incorrect document-to-section mapping leads to generated content that cites wrong sources or misses critical data.

Our Contribution: We present a hierarchical neural network classifier that: (1) Maps pharmaceutical documents to CTD sections with 74.9% accuracy, (2) Achieves near-perfect module-level classification (98.4%), (3) Outperforms TF-IDF baselines by 7.1 percentage points, (4) Enables automated document routing for LLM-based dossier generation.

2. Related Work

Document Classification: Transformer-based models have achieved state-of-the-art results on document classification tasks. BERT and its variants provide contextual embeddings that capture semantic meaning beyond simple keyword matching.

Biomedical NLP: BioBERT (Lee et al., 2020) pre-trained on PubMed abstracts has shown superior performance on biomedical text mining tasks, including named entity recognition and relation extraction.

Hierarchical Classification: Multi-level classification approaches exploit label hierarchies to improve predictions, particularly for fine-grained categories with limited training data.

3. Dataset

3.1 Data Source

We use documents from an Apixaban (anticoagulant drug) regulatory dossier, including: Clinical literature references (Module 2), Quality/CMC documentation (Module 3), and Clinical study reports (Module 5).

3.2 Statistics

Metric	Value
Total Documents	912
CTD Sections (classes)	10
CTD Modules	3
Train/Test Split	729/183 (80/20)

3.3 Class Distribution

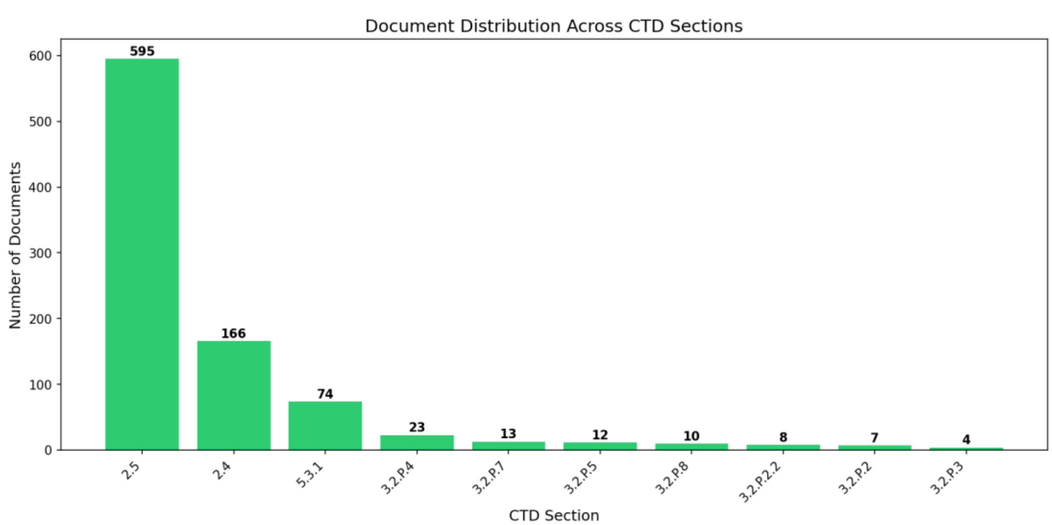

Figure 1: Document distribution across CTD sections

The dataset exhibits significant class imbalance, with clinical literature (Sections 2.4 and 2.5) comprising 83% of documents.

4. Methods

4.1 Document Representation

We extract text from PDF and Word documents using PyMuPDF and python-docx. Each document is truncated to 10,000 characters. Documents are embedded using BioBERT (dmis-lab/biobert-base-cased-v1.2), producing 768-dimensional vectors that capture biomedical semantic content.

4.2 Hierarchical Architecture

Our model employs a two-level hierarchical approach: Level 1 predicts the CTD module (2, 3, or 5) with 3 output classes. Level 2 predicts the specific CTD section with 10 output classes, conditioned on Level 1 predictions.

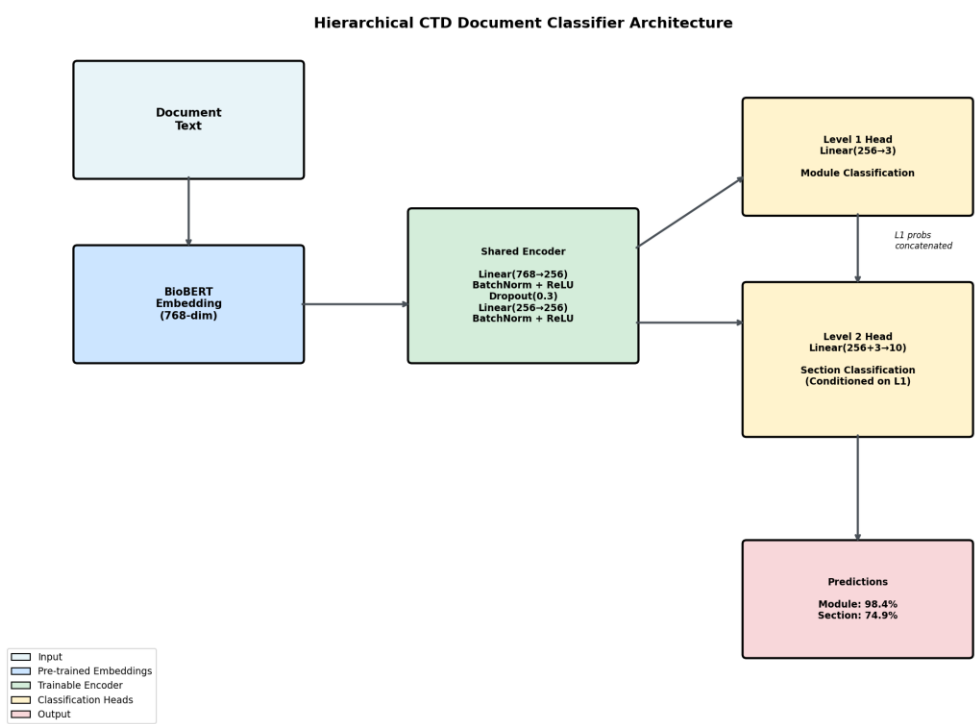

Figure 2: Hierarchical classifier architecture

The architecture consists of:

• Shared Encoder: Linear(768→256) + BatchNorm + ReLU + Dropout(0.3) × 2 layers

• Level 1 Head: Linear(256→3) for module classification

• Level 2 Head: Linear(256+3→10), concatenates shared features with Level 1 probabilities

4.3 Hierarchical Loss Function

We use a weighted combination: $L = 0.3 \times L\_level1 + 0.7 \times L\_level2$

The higher weight on Level 2 emphasizes fine-grained classification while benefiting from hierarchical structure.

4.4 Training Details

Hyperparameter	Value
Optimizer	Adam
Learning Rate	1e-3
Weight Decay	1e-5
Batch Size	16
Epochs	50
Dropout	0.3
LR Scheduler	ReduceLROnPlateau (patience=5)

5. Results

5.1 Main Results

Method	Accuracy	Improvement
TF-IDF + Logistic Regression	67.76%	baseline
BioBERT + MLP	68.31%	+0.6%
BioBERT + Logistic Regression	69.95%	+2.2%
BERT (general) + LogReg	71.58%	+3.8%
BioBERT + SVM	73.77%	+6.0%
Hierarchical NN (Ours)	74.86%	+7.1%

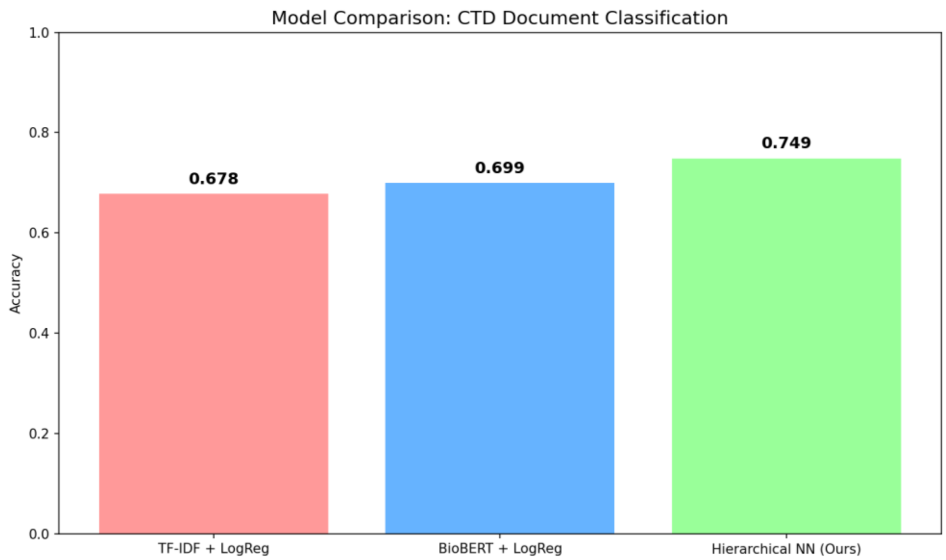

Figure 3: Model comparison showing accuracy improvements

5.2 Hierarchical Performance

• Module Level (3 classes): 98.36% accuracy

• Section Level (10 classes): 74.86% accuracy

The near-perfect module-level accuracy demonstrates the effectiveness of the hierarchical approach.

5.3 Training Dynamics

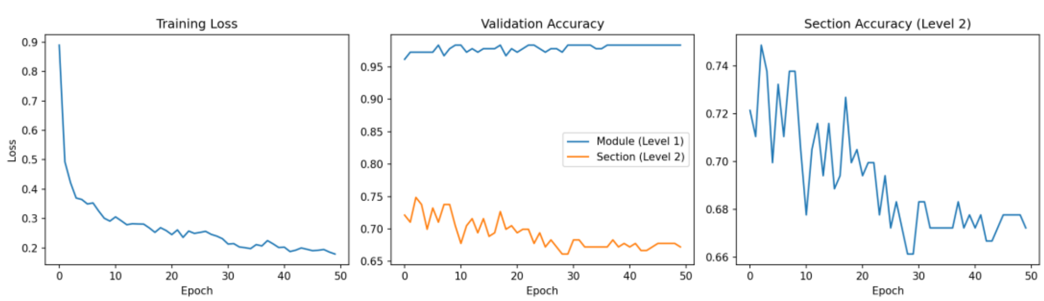

Figure 4: Training loss and validation accuracy over epochs

5.4 Confusion Matrix

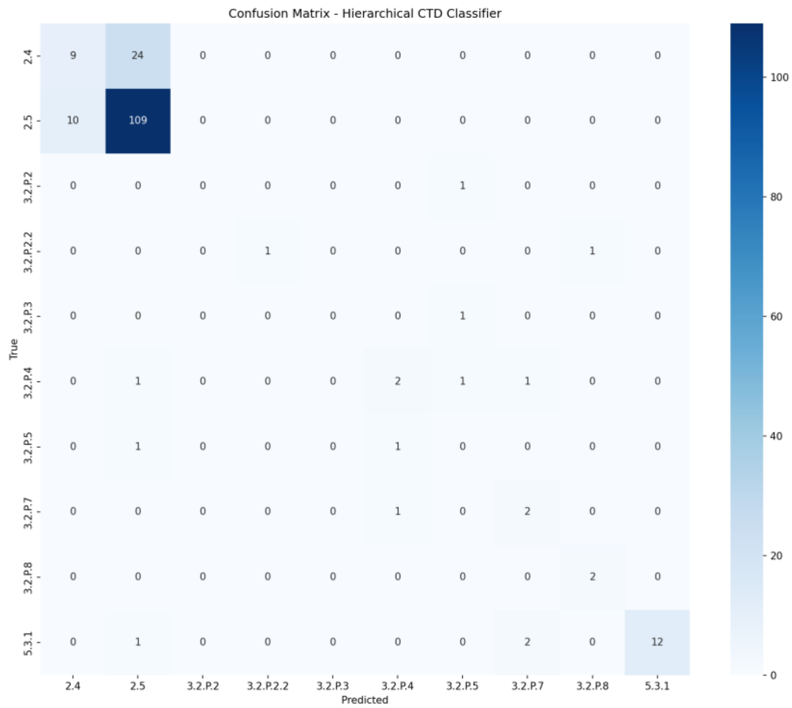

Figure 5: Confusion matrix for section-level predictions

The model performs well on sections with sufficient training data (2.5, 5.3.1) but struggles with extremely rare classes (3.2.P.2, 3.2.P.3) that have fewer than 5 training examples.

5.5 Ablation Studies

Effect of Classifier Architecture:

Classifier	Accuracy
Logistic Regression	69.95%
MLP (256, 128)	68.31%
SVM (RBF)	73.77%
Hierarchical NN	74.86%

6. Discussion

6.1 Key Findings

1. Hierarchical classification improves performance by first predicting the module (98.4% accuracy), providing context for section classification.

2. Deep learning outperforms traditional methods with 7.1% improvement over TF-IDF baseline.

3. Class imbalance remains challenging for sections with fewer than 5 training examples.

6.2 Limitations

• Small dataset: 912 documents from single drug product limits generalization

• Class imbalance: 65% of data belongs to one class (2.5)

• PDF extraction quality: Some documents yield poor text extraction

6.3 Future Work

1. Data augmentation for rare classes

2. Multi-dossier training across drug products

3. Fine-tuning BioBERT end-to-end

4. Production deployment at SagaReg

7. Conclusion

We presented a hierarchical deep learning approach for classifying pharmaceutical regulatory documents into CTD sections. Our model achieves 74.9% accuracy, outperforming TF-IDF baselines by 7.1 percentage points. The hierarchical structure proves particularly effective, achieving 98.4% accuracy at the module level. This classifier enables automated document routing for LLM-powered dossier generation.

References

1. Lee, J., et al. (2020). BioBERT: a pre-trained biomedical language representation model. Bioinformatics.

2. Devlin, J., et al. (2019). BERT: Pre-training of Deep Bidirectional Transformers. NAACL.

3. ICH M4 (2016). Organisation of the Common Technical Document.